



**VIT-AP**  
**UNIVERSITY**

# **Fundamentals of Electrical and Electronics Engineering (FEEE)**

**SUB CODE: ECE1002**

**T-P-C: 3-2-4**

**Lecture-34**

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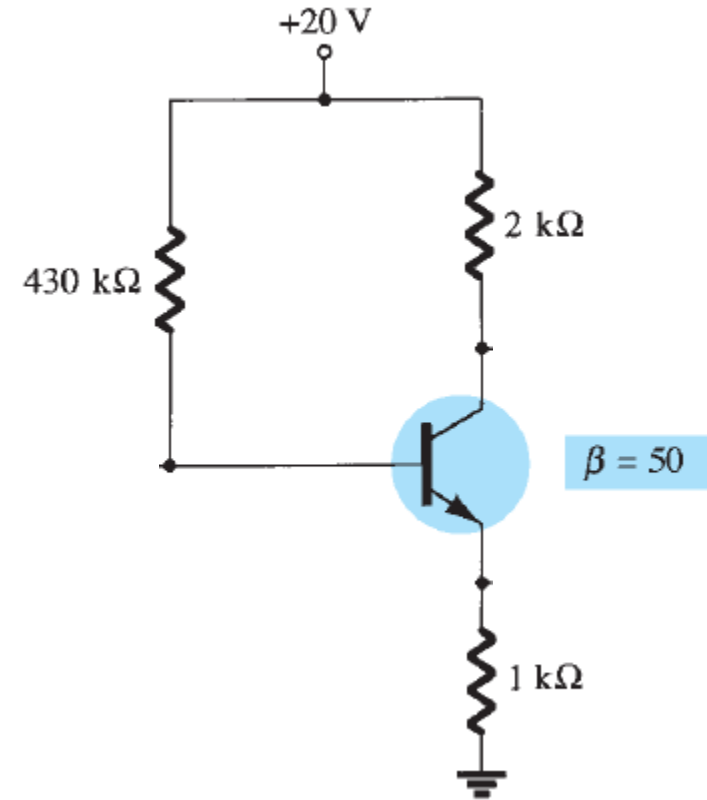
**Cabin: Faculty Area 424-C:AB-1**

# Transistor Model Problems

# Problem

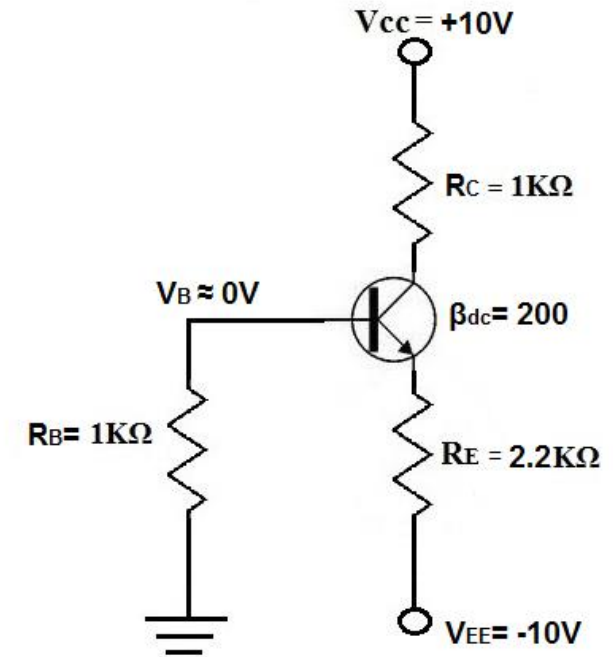
- For the emitter-bias network of Fig. below, determine:

- a.  $I_B$
- b.  $I_C$
- c.  $V_{CE}$
- d.  $V_C$
- e.  $V_E$
- f.  $V_B$
- g.  $V_{BC}$



# Problem

- Calculate the emitter current  $I_E$  and collector voltage  $V_C$ .



# Problem

- For the fixed bias with emitter resistance with  $R_c = 1\text{K}\Omega$ ,  $R_b = 2\text{K}\Omega$  and  $R_E = 1\text{K}\Omega$ .  $V_{cc} = -15\text{ v}$ ,  $V_{BB} = 5\text{v}$  and  $V_{EE} = -10\text{ v}$ . Find following parameters. ( $\beta = 50$ )
  - a.  $I_B$
  - b.  $I_C$  and  $I_E$
  - c.  $V_{CE}$
  - d.  $V_C$
  - e.  $V_E$
  - f.  $V_B$
  - g.  $V_{BC}$

# Problem

For the given emitter bias circuit, find the following parameters.

- i)  $R_C$
- ii)  $R_E$
- iii)  $R_B$
- iv)  $V_{CE}$
- v)  $V_B$

